

Assignment 9

ROLL NO:CSB24083

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import java.util.*;

class Account {
    private String accountNumber;
    private String ownerName;
    private double balance;

    public Account() {
        this("0000", "Unknown", 0.0);
    }

    public Account(String accountNumber, String ownerName, double
balance) {
        this.accountNumber = accountNumber;
        this.ownerName = ownerName;
        if (balance < 0) {
            throw new IllegalArgumentException("Initial balance
cannot be negative");
        }
        this.balance = balance;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public String getOwnerName() {
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        return ownerName;
    }

    public double getBalance() {
        return balance;
    }

    public void setOwnerName(String ownerName) {
        this.ownerName = ownerName;
    }

    protected void setBalance(double balance) {
        this.balance = balance;
    }

    public void deposit(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Deposit amount
must be positive");
        }
        balance += amount;
        System.out.println("Deposited: " + amount);
    }

    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Withdraw amount
must be positive");
        }
        if (amount > balance) {
            throw new IllegalArgumentException("Insufficient
balance");
        }
        balance -= amount;
        System.out.println("Withdrawn: " + amount);
    }
}
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        public void display() {
            System.out.println("Account Number : " + accountNumber);
            System.out.println("Owner Name      : " + ownerName);
            System.out.printf("Balance          : %.2f\n", balance);
        }
    }

class SavingsAccount extends Account {
    private double interestRate;

    public SavingsAccount(String accountNumber, String ownerName,
double balance, double interestRate) {
        super(accountNumber, ownerName, balance);
        if (interestRate < 0) {
            throw new IllegalArgumentException("Interest rate
cannot be negative");
        }
        this.interestRate = interestRate;
    }

    public double calculateInterest() {
        return getBalance() * interestRate / 100;
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Account Type    : Savings");
        System.out.println("Interest Rate   : " + interestRate +
"%");
        System.out.printf("Interest Earned: %.2f\n",
calculateInterest());
    }
}

class CurrentAccount extends Account {
    private double overdraftLimit;

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        public CurrentAccount(String accountNumber, String ownerName,
double balance, double overdraftLimit) {
            super(accountNumber, ownerName, balance);
            if (overdraftLimit < 0) {
                throw new IllegalArgumentException("Overdraft limit
cannot be negative");
            }
            this.overdraftLimit = overdraftLimit;
        }

        @Override
        public void withdraw(double amount) {
            if (amount <= 0) {
                throw new IllegalArgumentException("Withdraw amount
must be positive");
            }
            if (amount > getBalance() + overdraftLimit) {
                throw new IllegalArgumentException("Overdraft limit
exceeded");
            }
            setBalance(getBalance() - amount);
            System.out.println("Withdrawn (with overdraft allowed): "
+ amount);
        }

        @Override
        public void display() {
            super.display();
            System.out.println("Account Type    : Current");
            System.out.printf("Overdraft Limit: %.2f\n",
overdraftLimit);
        }
    }

    public class BankSystem {
        public static void main(String[] args) {

```

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        List<Account> accounts = new ArrayList<>();

        accounts.add(new SavingsAccount("101", "Alice", 5000,
5));
        accounts.add(new CurrentAccount("102", "Bob", 2000,
1000));

        System.out.println("---- Performing Transactions ----");
        accounts.get(0).deposit(1000);
        accounts.get(1).withdraw(2500);

        System.out.println("\n---- Account Details ----");
        for (Account acc : accounts) {
            System.out.println();
            acc.display();
        }
    }
}

```

Output:

---- Performing Transactions ----

Deposited: 1000.0

Withdrawn (with overdraft allowed): 2500.0

---- Account Details ----

Account Number : 101

Owner Name : Alice

Balance : 6000.00

Account Type : Savings

Interest Rate : 5.0%

Interest Earned: 300.00

Account Number : 102

Owner Name : Bob

Balance : -500.00

Account Type : Current

Overdraft Limit: 1000.00